

Air Quality-Based Health Risk Classification with Machine Learning Models

Introduction

Air pollution is a critical global health issue, responsible for millions of deaths each year. In Thailand, particularly during dry seasons and in the big cities, citizens regularly face dangerous air quality levels due to PM2.5 pollution. As someone from Thailand, I have experienced firsthand how poor air quality can affect daily life, from respiratory irritation to public health warnings telling people to stay indoors. Air is essential to life, yet in many parts of the world, just the simple breathing comes with hidden risks. This realisation motivates the need for systems that can help assess and predict air-related health risks more effectively.

This project investigates the use of machine learning to predict the health impact class, a categorical label reflecting overall health risk. The dataset used in this study, Air Quality and Health Impact Data, contains 5,811 records with detailed information on air pollutant levels, weather conditions, and health metrics. The data was made available for educational use by Rabie El Kharoua on Kaggle [1]. The aim is to build a model capable of predicting health risk levels from environmental inputs by applying classification algorithms to this data. Such a model could support early public health interventions and increase awareness about the dangers of poor air quality.

Literature review

The dataset used in this study includes multiple pollutants and weather conditions, along with health outcomes, allowing for the exploration of complex relationships between air quality and public health. The target variable, health impact class, is derived from a severity score and categorised into five levels from very low to very high. This type of data is suitable for classification tasks and aligns with existing research that uses air quality metrics to forecast health outcomes.

Numerous studies have established strong associations between air pollution and health conditions. Bellinger et al. [2] provided a systematic review of machine learning and data mining applications in air pollution epidemiology. Their findings emphasised the limitations of traditional epidemiological models and highlighted the potential of machine learning approaches, including neural networks, support vector machines, and decision trees, to better handle large, heterogeneous environmental health datasets. The review concluded that machine learning techniques could significantly enhance prediction accuracy and feature extraction. Similarly, Masih [3] conducted a review of 38 studies applying machine learning to environmental modelling and identified two primary tasks, which are pollutant estimation and pollution forecasting. The study found that ensemble learning, and linear regression methods were most effective for estimation tasks, while neural networks and support vector machines were commonly used for forecasting. Among the methods analysed, ensemble learning models have the highest average correlation coefficients and demonstrate more stable performance than other algorithms. These findings support the use of ensemble models, such as random forest, in air quality-related classification tasks.

This project builds on those findings by using classification algorithms to predict air pollution-related health impact categories, offering a potential decision-support tool for public health policy.

Methods

1. *Feature Construction and Preprocessing*

To prepare the data for classification, 55 features from the original dataset were constructed, which contained variables related to air quality, weather conditions, and public health. Feature construction included pollutant ratios, aggregated metrics, interaction terms, for example, the interaction between temperature and humidity, and polynomial and logarithmic transformations, e.g. log of PM2.5, squared AQI. Health-related ratios were also added, such as respiratory cases per AQI. Continuous features like temperature, humidity, and AQI were categorised into categorical levels and one-hot encoded. To avoid data leakage, the continuous health impact score variable, which directly determines the class label, was excluded. Only the target health impact class was retained for classification.

The dataset was examined for missing values, and none were found. Numerical features were scaled using z-score normalisation to ensure equal weighting in sensitive models, such as logistic regression. Class imbalance was addressed by applying balanced class weights in all classification algorithms. The weighting assigns higher importance to minority classes during training and helps mitigate bias toward the majority class.

2. *Machine Learning Algorithms*

Tree classification models were selected for a range of learning capacities and bias-variance trade-offs. Logistic Regression was used as a high-bias, linear baseline. The decision tree was chosen for its ability to model non-linear relationships with low bias and high variance. Random forest, an ensemble of decision trees, was selected for its robustness, improved generalisation, and performance on tabular data. All models were implemented using scikit-learn and trained on the full set of constructed and pre-processed features. Class balancing was applied to all models to ensure a fair comparison under the dataset's imbalanced conditions.

3. *Cross-validation tuning*

To ensure unbiased evaluation, nested cross-validation was applied. The outer loop used 5-fold stratified cross-validation to evaluate model performance on unseen data, while the inner loop used 3-fold cross-validation for hyperparameter tuning. This approach prevents information leakage from the validation set into the training process and offers a reliable estimate of each model's generalisation ability. All three models were evaluated using the same nested setup, ensuring consistent comparison. The macro-averaged precision, recall, F1-score, and accuracy were used as evaluation metrics to measure the model's performance across all classes, regardless of class imbalance.

4. *Hyperparameter tuning*

Each model underwent hyperparameter tuning via GridSearchCV within the inner folds of the nested cross-validation loop. For logistic regression, the regularisation strength parameter C over values [0.1, 1, 10], covering a wide spectrum of regularisation, were used. For the decision tree model, max depth values of 5, 10, and 15 were tested to control tree complexity and prevent overfitting. For Random Forest, the number of trees n_estimators was tuned using values of 50, 100, and 200.

Result and discussion

To evaluate model performance, nested cross-validation with a 5-fold outer loop and a 3-fold inner loop for hyperparameter tuning was used. Each model was assessed using accuracy, precision, recall, and F1-score. The table below presents the results across all folds.

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	0.7169	0.3682	0.5270	0.3923
Decision Tree	0.8931	0.5402	0.5944	0.5571
Random Forest	0.9184	0.6654	0.4836	0.5305

Table 1: Performance of classification models on health impact prediction

Among the three models, the random forest model achieved the highest overall accuracy (91.8%), indicating a strong capability to predict, particularly for the dominant classes in the dataset. However, its lower recall (48.4%) and moderate F1-score (53.1%) suggest that minority classes, representing higher health risks, are still under-predicted. The decision tree classifier achieved slightly lower accuracy (89.3%) but demonstrated the highest F1-score (55.7%), indicating more balanced performance across all classes. This may be due to its ability to model non-linear interactions without relying on ensemble averaging, making it better at capturing patterns across underrepresented classes. The logistic regression, while simple and interpretable, underperformed across all metrics, particularly with a precision of 36.8% and F1-score of 39.2%. This is likely due to its linear nature and unable to capture the complex relationships in the dataset. Overall, while random forest offers strong performance for high-confidence predictions, the decision tree's balanced recall and F1-score suggest it may be more suitable in settings where false negatives carry a higher cost.

Conclusion

This project explored the application of machine learning classification algorithms to predict the health impact class from air quality and environmental data. By constructing 55 derived features from the original dataset and applying standard preprocessing techniques such as scaling and class rebalancing, able to develop and evaluate three models using a nested cross-validation.

The results show that ensemble methods like random forest offer high overall accuracy, but sacrifice recall on minority classes. In contrast, the decision tree model provided a more balanced performance across all metrics, making it a strong candidate for practical use in risk-sensitive public health applications. Logistic regression, while simple, was limited in handling the nonlinear and imbalanced data.

In the future, the model's performance could be improved by using better sampling techniques to handle class imbalance or by trying more advanced models like gradient boosting. Overall, this project shows that machine learning can help understand how air quality affects health and can be a useful starting point for building tools to support air pollution monitoring and public health decisions.

References

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